MeEn537 Final Project Proposal

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Project Description

We will use Baxter's integrated camera to generate a 3-D map of an object. This will include capturing images with the camera placed an known points, and developing an algorithm to stich the images together into a 3-D map.

Success Measures

- Capture images of a subject with the camera at a known location using Baxter's integrated camera or our own camera mounted to Baxter
- Develop an algorithm that can construct a recognizable 3-D map of the subject using 2 captured images

Stretch Goals

- Implement support for more than 2 images

Potential Pitfalls

- Camera lens distortion
- Insufficient camera quality from Baxter's integrated camera

Timeline

Date	Task
2026-11-06	Project proposal
2026-11-09	Check image quality
2026-11-11	Correct lens distortion
2026-11-13	Initial image capture
2026-11-20	Implement feature detection
2026-11-25	Triangulate location of points
2026-11-26	Render point cloud
2026-12-04	Improve reconstruction, add additional viewpoints

Resources and Libraries

We will use Python libraries such as OpenCV and use the book *Robotics, Vision and Control* by Peter Corke as a reference.